

HR Data Analysis

Employee Attrition and Retention Analysis (Workforce Insights)

SQL + Power BI | Data-Driven HR Analytics



Problem Statement

Organizations often face challenges with high employee attrition, leading to increased recruitment costs, reduced productivity, and workforce instability. This project aims to help HR departments analyze employee data using SQL, Power BI, and Power Query to identify the key factors influencing employee turnover, such as salary, job role, experience, education field, and demographics. Through interactive dashboards and analytical insights, the solution supports data-driven HR decision-making for improving employee retention, workforce planning, and overall organizational performance.

Business Impact (HR-Focused & Actionable)

- Which employee groups are most likely to leave?
- Which factors combined drive early attrition?
- Where should retention budgets be prioritized?
- How can we identify at-risk employees before they resign?
- What role do promotion cycles play in turnover?

Attrition rate by Gender

```
1  -- Attrition rate by Gender
2  •  SELECT
3      Gender,
4      COUNT(*) AS Total_Employees,
5      SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrition_Count,
6      ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate_Pct
7  FROM hr_analysis_new
8  GROUP BY Gender;
```

Result Grid				
Filter Rows:		Export:	Wrap Cell Content:	
Gender	Total_Employees	Attrition_Count	Attrition_Rate_Pct	
Female	588	87	14.80	
Male	882	150	17.01	

Attrition by Over Time

Larana, Inc.

Process

Tools

Applications

```
1  -- Attrition by Over Time
2  •  select
3      OverTime,
4          COUNT(*) AS Total,
5          SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
6          ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
7  FROM hr_analysis_new
8  GROUP BY OverTime;
```

result Grid			Filter Rows:		Export:		Wrap Cell Content:	
OverTime	Total	Attrited	Attrition_Rate					
No	1054	110	10.44					
Yes	416	127	30.53					

Department-wise attrition rate with ranking

```
1  -- Department-wise attrition rate with ranking
2
3  •  SELECT
4      Department,
5      COUNT(*) AS Total,
6      SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
7      ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate,
8      RANK() OVER (ORDER BY ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) DESC) AS Risk_Rank
9  FROM hr_analysis_new
10 GROUP BY Department;
```

result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

Department	Total	Attrited	Attrition_Rate	Risk_Rank
Sales	446	92	20.63	1
Human Resources	63	12	19.05	2
Research & Development	961	133	13.84	3

Top 5 job role by highest attrition count

```
1 SELECT
2     JobRole,
3     COUNT(*) AS Attrition_Count
4 FROM hr_analysis_new
5 WHERE Attrition = 'Yes'
6 GROUP BY JobRole
7 ORDER BY Attrition_Count DESC
8 LIMIT 5;
```

Result Grid		Filter Rows:	
JobRole	Attrition_Count		
Laboratory Technician	62		
Sales Executive	57		
Research Scientist	47		
Sales Representative	33		
Human Resources	12		



Attrition rate by job satisfaction level (1-4)

Lorand, Inc.

Process

Tools

Applications

```
1  -- Attrition rate by job satisfaction level (1-4)
2
3  •  select
4      JobSatisfaction,
5          ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
6  FROM hr_analysis_new
7  GROUP BY JobSatisfaction
8  ORDER BY JobSatisfaction;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	JobSatisfaction	Attrition_Rate
	1	22.84
	2	16.43
	3	16.52
	4	11.33

Job role + Overtime combination with highest attrition

```
1  -- Job role + Overtime combination with highest attrition
2  •  SELECT
3      JobRole,
4      OverTime,
5      COUNT(*) AS Attrition_Count
6  FROM hr_analysis_new
7  WHERE Attrition = 'Yes'
8  GROUP BY JobRole, OverTime
9  ORDER BY Attrition_Count DESC
10 LIMIT 10;
```

JobRole	OverTime	Attrition_Count
Research Scientist	Yes	33
Laboratory Technician	No	31
Laboratory Technician	Yes	31
Sales Executive	Yes	31
Sales Executive	No	26
Sales Representative	No	17
Sales Representative	Yes	16
Research Scientist	No	14
Human Resources	No	7
Healthcare Representative	No	7



Average years at company for attrited vs active

Applications

```
1  -- Average years at company for attrited vs active
2
3  •  SELECT
4      Attrition,
5      ROUND(AVG(YearsAtCompany), 2) AS Avg_Years_At_Company
6  FROM hr_analysis_new
7  GROUP BY Attrition;
```

Result Grid

  Filter Rows:

Export: 

Wrap Cell Content: 

Attrition	Avg_Years_At_Company
No	7.37
Yes	5.13

Attrition by Enviorment Satisfaction

```
1  -- Attrition by Enviorment Satisfaction
2
3  •  SELECT
4      EnvironmentSatisfaction,
5      COUNT(*) AS Total,
6      SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
7      ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
8  FROM hr_analysis_new
9  GROUP BY EnvironmentSatisfaction
10 ORDER BY EnvironmentSatisfaction;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 				
	EnvironmentSatisfaction	Total	Attrited	Attrition_Rate
1		284	72	25.35
2		287	43	14.98
3		453	62	13.69
4		446	60	13.45

Attrition by Years Since Last Promotion

```
1  -- Attrition by Years Since Last Promotion
2
3  ●  SELECT
4      YearsSinceLastPromotion,
5      COUNT(*) AS Attrited_Count
6  FROM hr_analysis_new
7  WHERE Attrition = 'Yes'
8  GROUP BY YearsSinceLastPromotion
9  ORDER BY Attrited_Count DESC
10  LIMIT 5;
```

Result Grid		Filter Rows:	Export:
YearsSinceLastPromotion	Attrited_Count		
0	110		
1	49		
2	27		
7	16		
3	9		



High Performer Attrition Risk

```
3 • SELECT
4     `EmpID`,`JobRole`,Department,`PerformanceRating`,`MonthlyIncome`,`YearsAtCompany`,
5     (
6         CASE WHEN `PerformanceRating` >= 4 THEN 3 ELSE 0 END +
7         CASE WHEN `JobSatisfaction` <= 2 THEN 3 ELSE 0 END +
8         CASE WHEN `PercentSalaryHike` < 13 THEN 2 ELSE 0 END +
9         CASE WHEN `OverTime` = 'Yes' THEN 2 ELSE 0 END +
10        CASE WHEN `YearsSinceLastPromotion` >= 3 THEN 1 ELSE 0 END
11    ) AS High_Performer_Exit_Risk
12 FROM hr_analysis_new
13 WHERE Attrition = 'No'
14 ORDER BY High_Performer_Exit_Risk desc
15 LIMIT 10;
```

EmpID	JobRole	Department	PerformanceRating	MonthlyIncome	YearsAtCompany	High_Performer_Exit_Risk
RM919	Manager	Sales	4	19847	29	9
RM278	Sales Executive	Sales	4	5605	8	9
RM883	Manufacturing Director	Research & Development	4	10252	7	9
RM613	Sales Executive	Sales	4	4779	8	9
RM887	Research Scientist	Research & Development	4	3579	11	9
RM452	Manufacturing Director	Research & Development	4	7406	10	9
RM1345	Research Scientist	Research & Development	4	4284	5	8
RM949	Manager	Research & Development	4	11916	9	8
RM264	Manager	Sales	3	16872	7	8
RM1468	Manufacturing Director	Research & Development	4	6142	6	8

Attrition trend by years at company (detailed)

Lorand, Inc.

```
1  -- Attrition trend by years at company (detailed)
2
3  • select
4      YearsAtCompany,
5      COUNT(*) AS Total_Employees,
6      SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
7      ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
8  FROM hr_analysis_new
9  GROUP BY YearsAtCompany
10 ORDER BY YearsAtCompany asc
11 limit 10;
12
```

Result Grid Filter Rows: Export: Wrap Cell Content: Fetch rows:				
	YearsAtCompany	Total_Employees	Attrited	Attrition_Rate
	0	44	16	36.36
	1	171	59	34.50
	2	127	27	21.26
	3	128	20	15.63
	4	110	19	17.27
	5	196	21	10.71
	6	76	9	11.84
	7	90	11	12.22
	8	80	9	11.25
	9	82	8	9.76
				11.25



Conclusion

The Employee Attrition and Retention Analysis project was developed to help HR teams identify the key reasons behind employee turnover and improve workforce retention. Using SQL, Power BI, and Power Query, workforce data was analyzed through business-driven SQL queries and interactive dashboards to uncover insights related to overtime, job role, experience, and employee satisfaction. The project supports data-driven HR decision-making by helping organizations reduce attrition, improve employee satisfaction, and optimize workforce planning.

Key HR Takeaways for Retention Strategy

- Sales & HR are crisis zones: They have the highest attrition rates (20.74% and 19.35%).
- Low satisfaction is a dealbreaker: Attrition rate jumps to 22.55% when job satisfaction is level 1.
- Pay is a baseline, not a solution: Most attrition happens in the lowest salary band.
- Watch your high-performers: Top talent is leaving due to low raises and lack of promotion.
- Overtime burns out critical roles: Especially high impact for Lab Technicians and Sales Executives.
- Stagnation predicts exit: If no promotion in 3+ years, attrition risk spikes.
- Target retention spend: Focus budget only on high performers, not all attrition.
- Flag men in specific roles: Male attrition count is higher and driven by specific jobs (Lab Tech, Sales).

Thank You



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